

DevOps Lab Assignment

Assignment TW 1.3 Basic Containerization (Docker) & Jenkins Freestyle Project

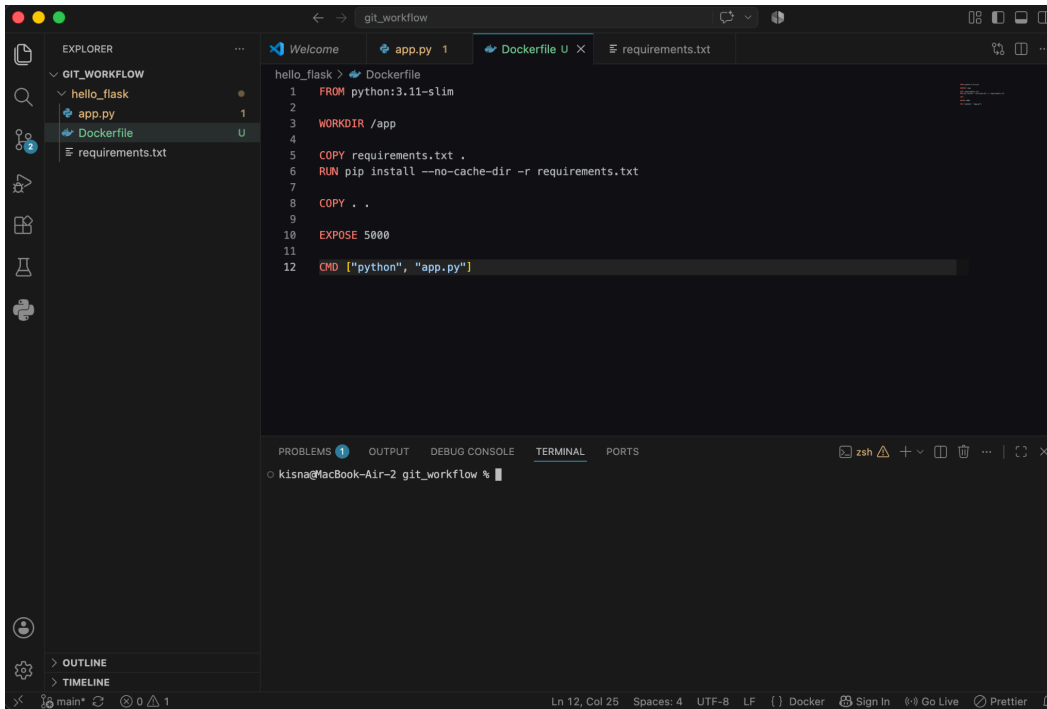
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Class : A3

Batch : 2023-27

Dockerfile



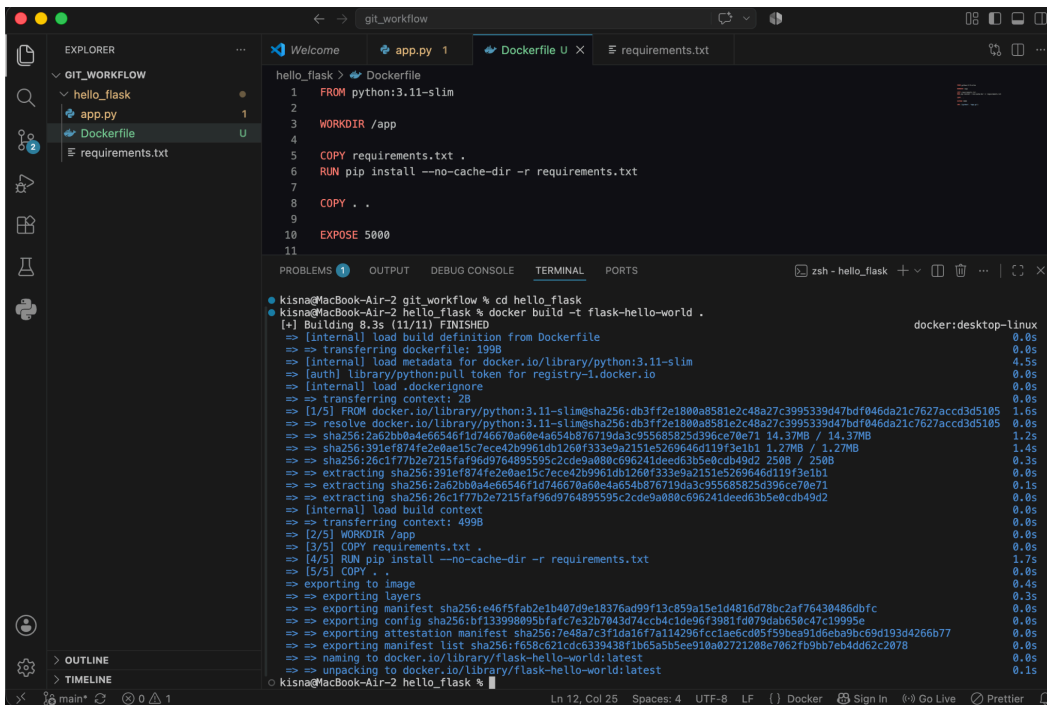
```
hello_flask > Dockerfile
1 FROM python:3.11-slim
2
3 WORKDIR /app
4
5 COPY requirements.txt .
6 RUN pip install --no-cache-dir -r requirements.txt
7
8 COPY . .
9
10 EXPOSE 5000
11
12 CMD ["python", "app.py"]
```

The screenshot shows a Visual Studio Code editor with a project named 'git_workflow'. The Explorer sidebar on the left shows the file structure: 'GIT_WORKFLOW' containing 'hello_flask', which includes 'app.py', 'Dockerfile', and 'requirements.txt'. The Dockerfile is open in the editor, showing the following content:

```
1 FROM python:3.11-slim
2
3 WORKDIR /app
4
5 COPY requirements.txt .
6 RUN pip install --no-cache-dir -r requirements.txt
7
8 COPY . .
9
10 EXPOSE 5000
11
12 CMD ["python", "app.py"]
```

The bottom panel shows the TERMINAL with the prompt 'kiswa@MacBook-Air-2 git_workflow %'.

Docker Build Output



```
hello_flask > Dockerfile
1 FROM python:3.11-slim
2
3 WORKDIR /app
4
5 COPY requirements.txt .
6 RUN pip install --no-cache-dir -r requirements.txt
7
8 COPY . .
9
10 EXPOSE 5000
11

kiswa@MacBook-Air-2 git_workflow % cd hello_flask
kiswa@MacBook-Air-2 hello_flask % docker build -t flask-hello-world .
[+] Building 8.3s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 199B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3d5185
=> resolve docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3d5185
=> sha256:2a62b08a4e65461d746870a60e4ad54b076719da3c95568582d396ce70e71 14.37MB / 14.37MB
=> sha256:391ef874fe2e8ae15c7ece42b9961db1260f333e9a2151e5269646d119f3e1b1 1.27MB / 1.27MB
=> sha256:26c1f77b2e7215faf96d9764895595c2cde9a808c696241deed63b5e0cd49d2 250B / 250B
=> extracting sha256:391ef874fe2e8ae15c7ece42b9961db1260f333e9a2151e5269646d119f3e1b1
=> extracting sha256:2a62b08a4e65461d746870a60e4ad54b076719da3c95568582d396ce70e71
=> extracting sha256:26c1f77b2e7215faf96d9764895595c2cde9a808c696241deed63b5e0cd49d2
=> [internal] load build context
=> transferring context: 499B
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> exporting layers
=> exporting manifest sha256:e46f5fab2e1b407d9e18376ad99f13c859a15e1d4816d78bc2af76438486dbfc
=> exporting config sha256:b133998095bafce32b7043d74ccb4c1de96f3981fd079dab658c47c19995e
=> exporting attestation manifest sha256:7e48a7c3f1da16f7a114296fcc1ae6c0d5f59bea91d6eba9bc69d193d4266b77
=> exporting manifest list sha256:f688e621cde6339438f1b65a5b5ee910a02721208e7062f99b7eb4dd62c2078
=> naming to docker.io/library/flask-hello-world:latest
=> unpacking to docker.io/library/flask-hello-world:latest
```

The screenshot shows the same Visual Studio Code editor as before, but the terminal now displays the output of the 'docker build' command. The output shows the build process for the 'flask-hello-world' image, including the transfer of the Dockerfile, loading of metadata, and the final build steps. The build is successful, and the image is named 'flask-hello-world:latest'.

Docker Image List

```
kisna@MacBook-Air-2 hello_flask % docker images
```

IMAGE	ID	DISK USAGE	Info →	U	In Use
285037554068.dkr.ecr.ap-south-1.amazonaws.com/transcodex-demo-api:latest	fbda6ee1c9a7	18.5MB			
285037554068.dkr.ecr.ap-south-1.amazonaws.com/transcodex-demo-dashboad:latest					
	54d59c5fb829	21MB			
285037554068.dkr.ecr.ap-south-1.amazonaws.com/transcodex-demo-demo:latest	4abdf4a0d326	21MB			
285037554068.dkr.ecr.ap-south-1.amazonaws.com/transcodex-demo-migrate:latest					
	60324da6468b	88MB			
285037554068.dkr.ecr.ap-south-1.amazonaws.com/transcodex-demo-reaper:latest					
	7349d49a53d4	10.9MB			
285037554068.dkr.ecr.ap-south-1.amazonaws.com/transcodex-demo-worker:latest					
	315d50d04020	222MB			
angular-app:latest	b8fa7c2b4f02	101MB			
angular-docker-angular-app:latest	f262f1b651e0	2.59GB			U
angular-docker-angular-prod:latest	eb221b83605d	101MB			
docker.n8n.io/n8nio/n8n:latest	3c07c723326d	2.5GB			U
docker:dind	66d292e5c26b	505MB			
elasticsearch:8.13.0	9d1cd1491778	1.26GB			U
flask-hello-world:latest	f658c621cdc6	236MB			

Docker Container Run

The screenshot shows a VS Code editor with a project named 'git_workflow'. The Explorer sidebar shows a folder 'hello_flask' containing 'app.py', 'Dockerfile', and 'requirements.txt'. The Dockerfile content is as follows:

```
1 FROM python:3.11-slim
2
3 WORKDIR /app
4
```

The terminal shows the command to run the container:

```
kisna@MacBook-Air-2 hello_flask % docker run -d -p 5002:5000 --name flask-container flask-hello-world
```

The output shows the container ID: `d9dc6315fc5411ced2f419a804e3408eabad49d9153c957c11b42e9dd809ee9f`.

Docker Container Status

The screenshot shows the same VS Code editor as before, but the terminal now displays the command to check the container status:

```
kisna@MacBook-Air-2 hello_flask % docker ps
```

The output shows the container is running:

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
d9dc6315fc54	flask-hello-world	"python app.py"	21 seconds ago	Up 20 seconds	0.0.0.0:5002->5000/tcp, [::]:5002->5000/tcp

Application Verification

The screenshot shows a web browser window with the address bar set to `localhost:5002`. The page content displays the message: `Hello from Main Branch!`

Jenkins Freestyle Project

GitHub Repository

kkbot122 / hello-flask

CodeIssuesPull requestsAgentsActionsProjectsWikiSecurity and qualityInsightsSettings

Files

main

Go to file

hello_flask

Dockerfile

app.py

requirements.txt

.DS_Store

hello-flask / hello_flask

kk bot and kk bot flask app7a6d53e · nowHistory

Name	Last commit message	Last commit date
..		
Dockerfile	flask app	now
app.py	flask app	now
requirements.txt	Initial Flask Hello World application	yesterday

Jenkins Dashboard

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

All

S

W

Name ↓

Last Success

Last Failure

Last Duration

✓☀️Declarative-pipeline-demo13 days#613 days#51.8 sec▶

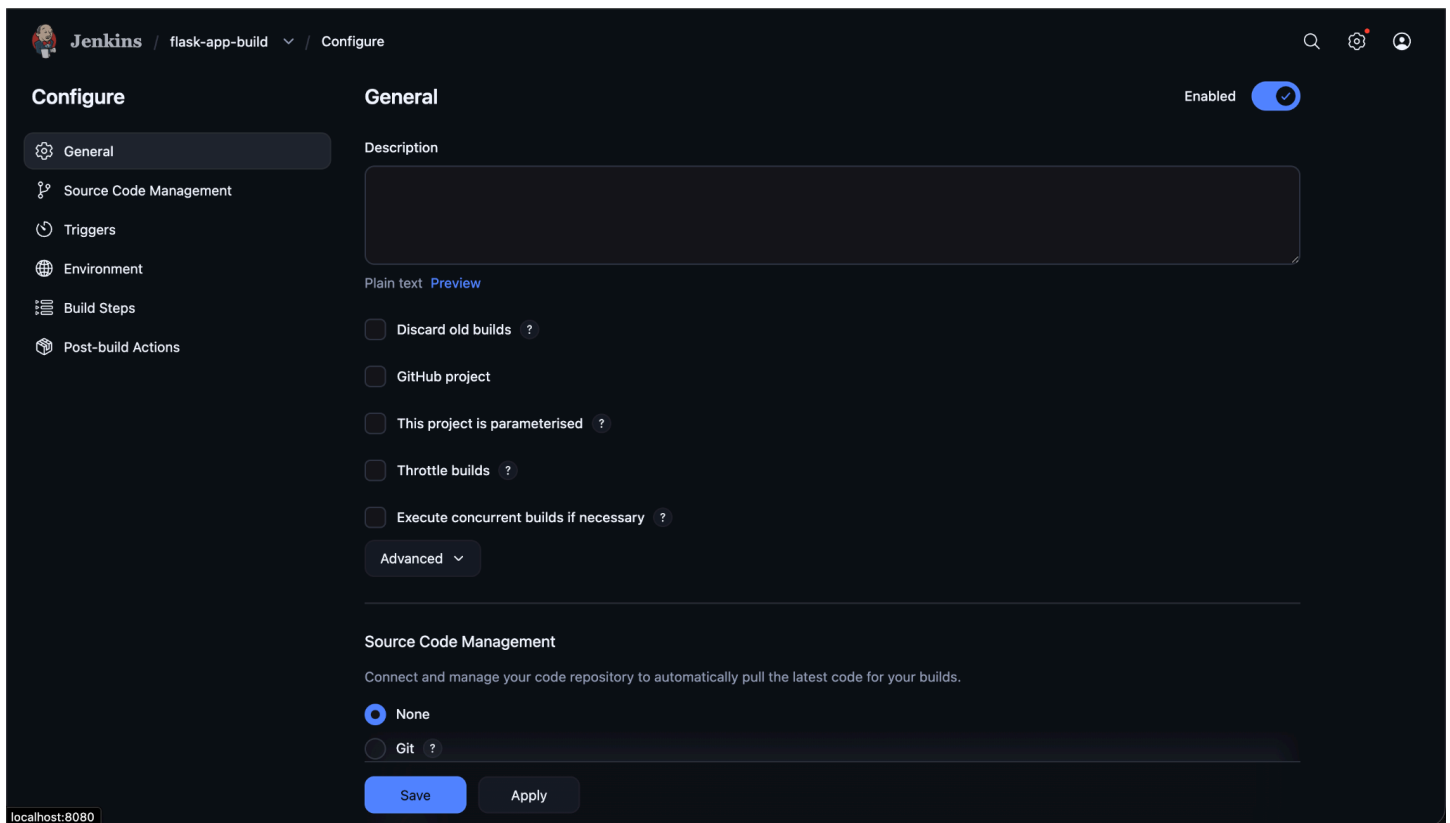
✓☁️Git-practice-build15 days#215 days#10.68 sec▶

Icon: SML

REST API

Jenkins 2.541.3

New Freestyle Project Creation



The screenshot shows the Jenkins configuration page for a new freestyle project named 'flask-app-build'. The 'General' tab is selected in the left sidebar. The main area is titled 'General' and includes a description field, a 'Plain text' preview, and several checkboxes for build options: 'Discard old builds', 'GitHub project', 'This project is parameterised', 'Throttle builds', and 'Execute concurrent builds if necessary'. An 'Advanced' dropdown is visible. Below these is the 'Source Code Management' section, which is currently set to 'None'. The 'Save' and 'Apply' buttons are at the bottom.

Jenkins / flask-app-build / Configure

Enabled ☒

General

Description

Plain text [Preview](#)

☐ Discard old builds ?

☐ GitHub project

☐ This project is parameterised ?

☐ Throttle builds ?

☐ Execute concurrent builds if necessary ?

Advanced ▾

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

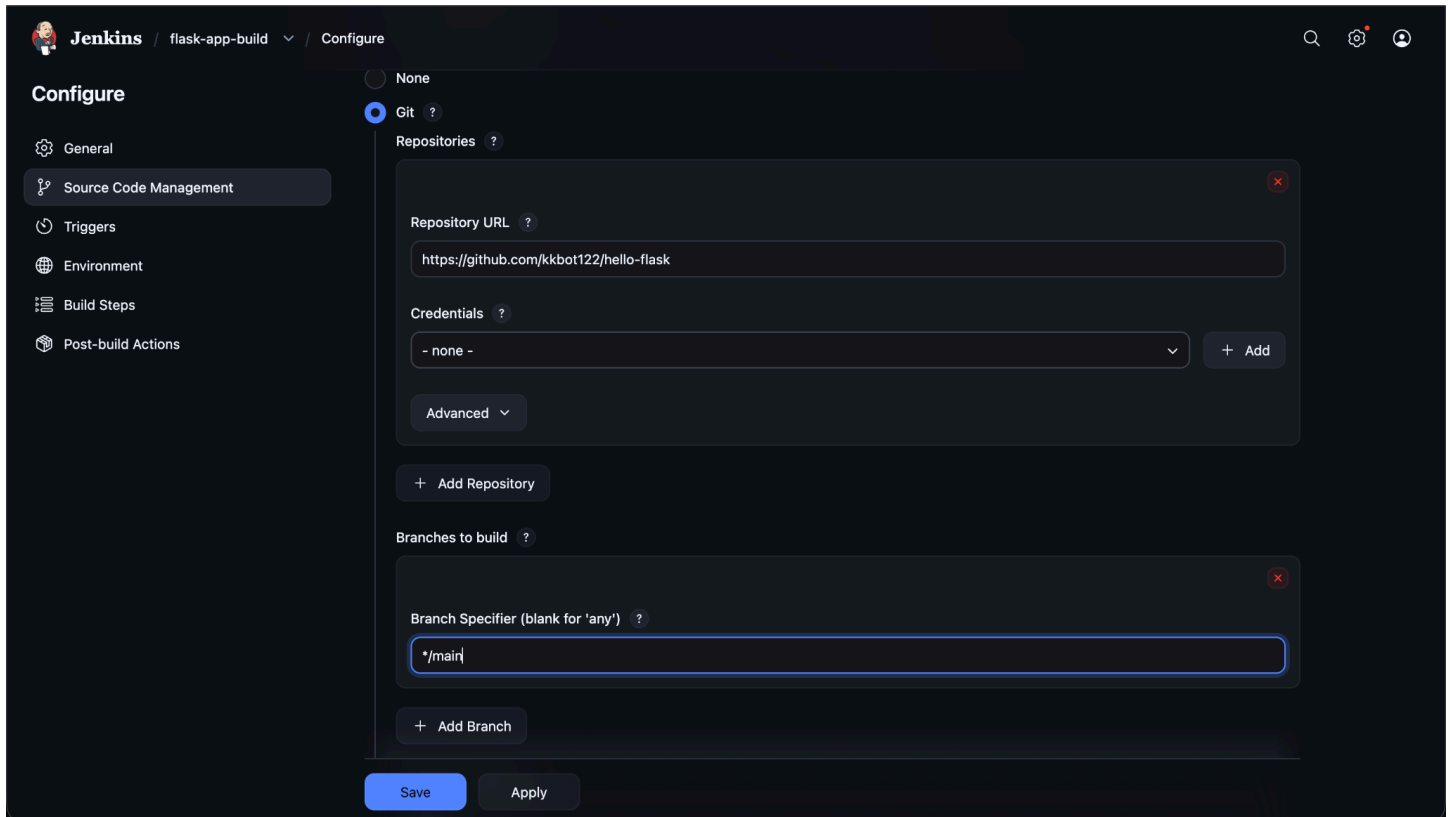
☒ None

☐ Git ?

Save Apply

localhost:8080

Source Code Management Configuration



The screenshot shows the Jenkins configuration page for the 'flask-app-build' project, now with the 'Source Code Management' tab selected. The 'Git' option is chosen under 'Source Code Management'. The 'Repositories' section contains a single repository with the URL 'https://github.com/kkbot122/hello-flask' and 'Credentials' set to '- none -'. The 'Branches to build' section has a 'Branch Specifier' set to '*/main'. The 'Save' and 'Apply' buttons are at the bottom.

Jenkins / flask-app-build / Configure

Configure

☐ None

☒ Git ?

Repositories ?

Repository URL ?

<https://github.com/kkbot122/hello-flask>

Credentials ?

- none - ▾ + Add

Advanced ▾

+ Add Repository

Branches to build ?

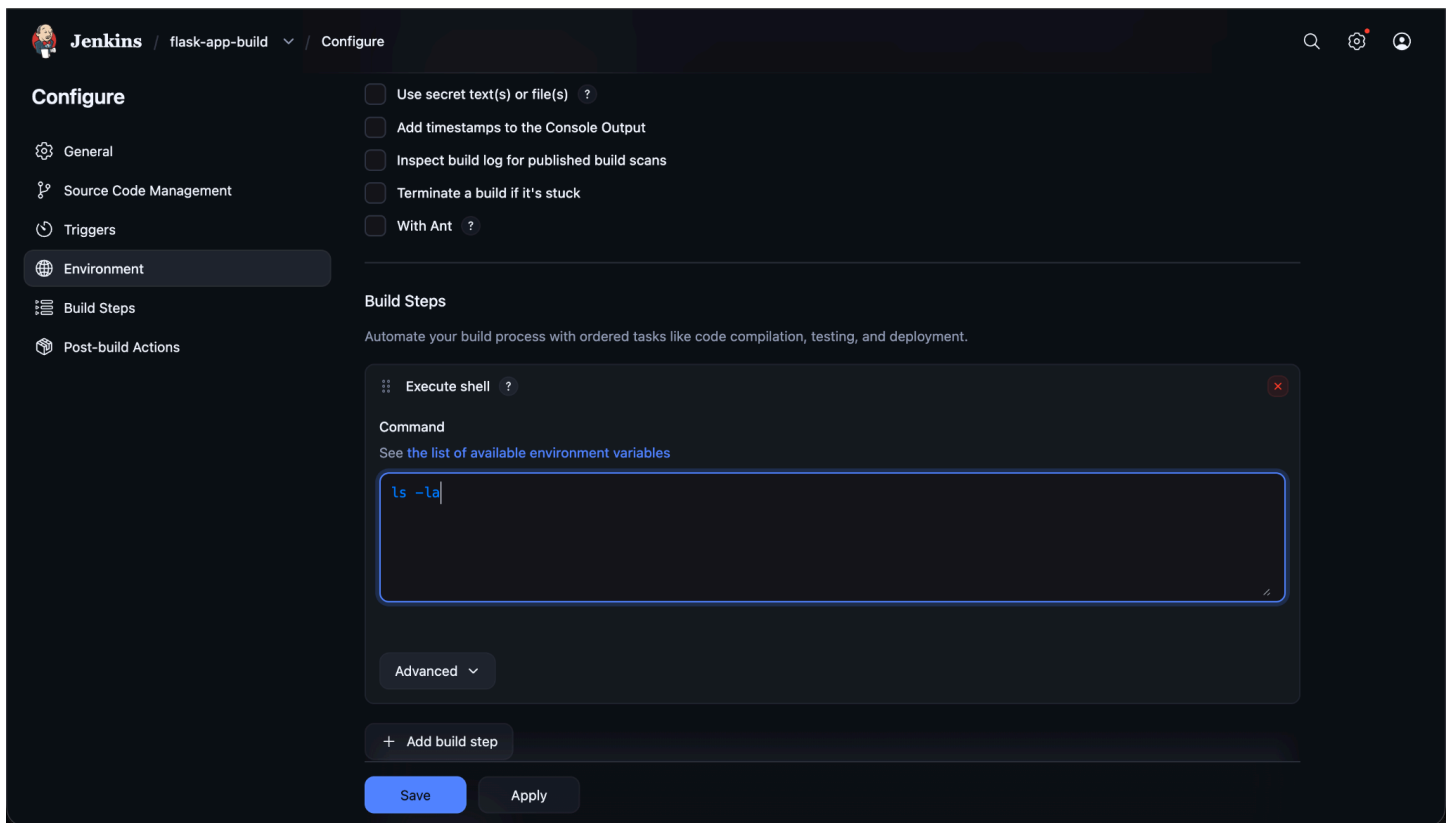
Branch Specifier (blank for 'any') ?

*/main

+ Add Branch

Save Apply

Build Step Configuration



This screenshot shows the Jenkins configuration page for a build step named 'flask-app-build'. The left sidebar contains navigation links: 'General', 'Source Code Management', 'Triggers', 'Environment' (which is selected), 'Build Steps', and 'Post-build Actions'. The main area is titled 'Configure' and contains several checkboxes for build options: 'Use secret text(s) or file(s)', 'Add timestamps to the Console Output', 'Inspect build log for published build scans', 'Terminate a build if it's stuck', and 'With Ant'. Below these is the 'Build Steps' section, which includes a description: 'Automate your build process with ordered tasks like code compilation, testing, and deployment.' A single build step is configured as 'Execute shell'. The 'Command' field contains the text 'ls -la'. Below the command field is an 'Advanced' dropdown menu. At the bottom of the configuration area are buttons for '+ Add build step', 'Save', and 'Apply'.

Jenkins / flask-app-build / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment**
- Build Steps
- Post-build Actions

☐ Use secret text(s) or file(s) ?

☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

☐ Terminate a build if it's stuck

☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See [the list of available environment variables](#)

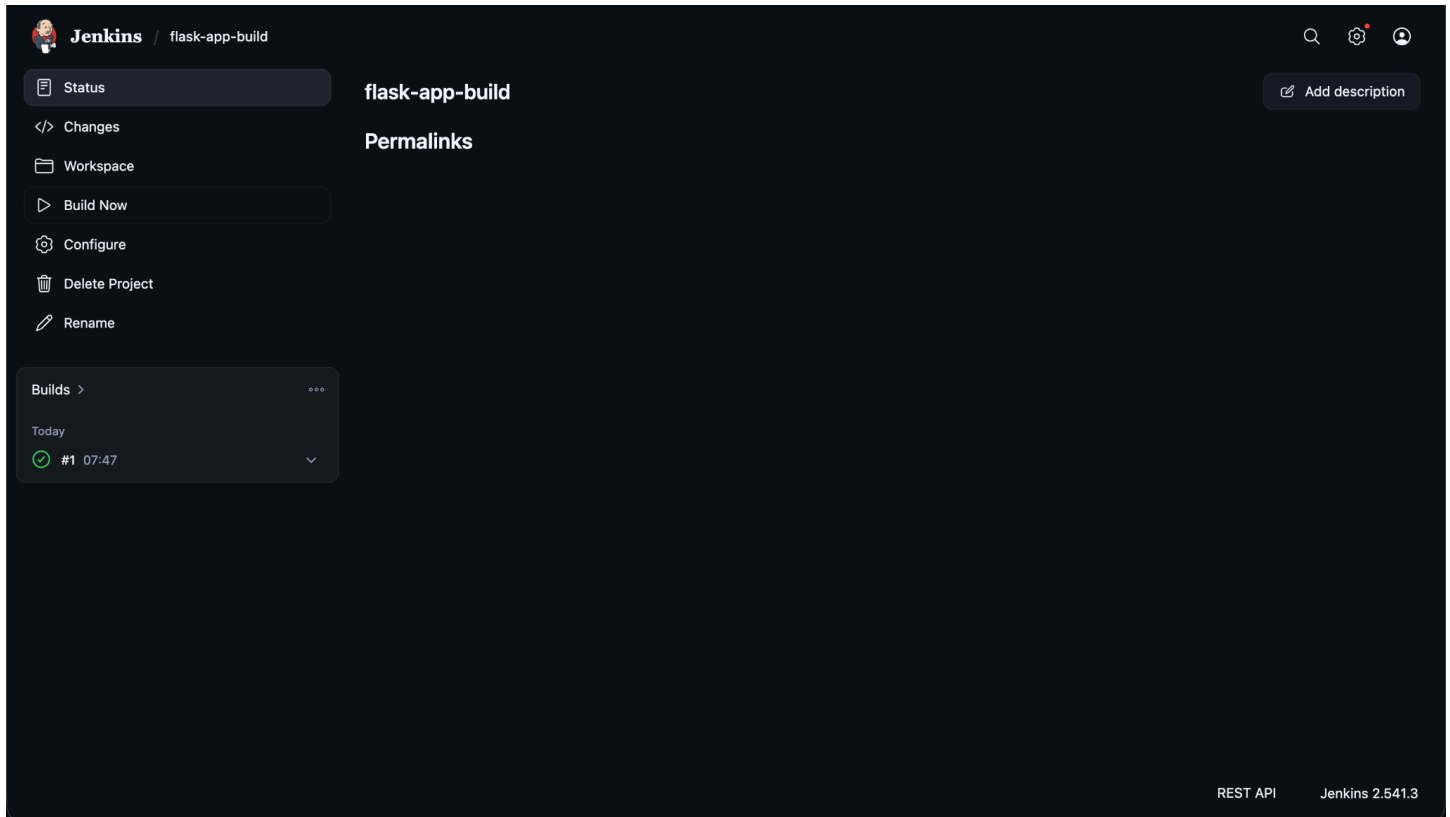
```
ls -la
```

Advanced ▾

+ Add build step

Save Apply

Build Triggered / Build History



This screenshot shows the Jenkins build history page for the 'flask-app-build' project. The left sidebar contains navigation links: 'Status' (selected), 'Changes', 'Workspace', 'Build Now', 'Configure', 'Delete Project', and 'Rename'. The main area is titled 'flask-app-build' and includes a link to 'Add description'. Below this is the 'Permalinks' section. The 'Builds' section shows a list of builds, with the first build being successful (indicated by a green checkmark) and labeled '#1 07:47'. The bottom of the page displays 'REST API' and 'Jenkins 2.541.3'.

Jenkins / flask-app-build

flask-app-build

[Add description](#)

Permalinks

Builds > ...

Today

- ☒ #1 07:47 ▾

REST API Jenkins 2.541.3

Console Output — Build Success

 **Jenkins** / flask-app-build ▾ / #1 ▾

Status

</> Changes

 Console Output

 Edit Build Information

 Delete build '#1'

 Git Build Data

Console

Download

Copy

View as plain text

```
Started by user Kisna Kanti
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/flask-app-build
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/kkbot122/hello-flask
> git init /var/jenkins_home/workspace/flask-app-build # timeout=10
Fetching upstream changes from https://github.com/kkbot122/hello-flask
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/kkbot122/hello-flask +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/kkbot122/hello-flask # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 7a6d53e7f3275e09efdc571c41da1126dbba64a5 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 7a6d53e7f3275e09efdc571c41da1126dbba64a5 # timeout=10
Commit message: "flask app"
First time build. Skipping changelog.
[flask-app-build] $ /bin/sh -xe /tmp/jenkins4838286019424002288.sh
+ ls -la
total 24
drwxr-xr-x 4 jenkins jenkins 4096 Jul 29 07:47 .
drwxr-xr-x 6 jenkins jenkins 4096 Jul 29 07:47 ..
-rw-r--r-- 1 jenkins jenkins 6148 Jul 29 07:47 .DS_Store
drwxr-xr-x 8 jenkins jenkins 4096 Jul 29 07:47 .git
drwxr-xr-x 2 jenkins jenkins 4096 Jul 29 07:47 hello_flask
Finished: SUCCESS
```

<https://github.com/kkbot122/hello-flask>